

Experiment 03

1. Experiment Title

Implement a Palindrome Checker using HTML, CSS and JavaScript.

2. Aim

To develop a web application that checks whether the entered word or sentence is a palindrome using JavaScript.

3. Theory

A palindrome is a word, phrase or sequence that reads the same forward and backward. JavaScript is used to reverse the entered string and compare it with the original string. DOM manipulation is used to read input and display the result dynamically.

4. Software/Tools Required

- HTML5
- CSS3
- JavaScript ES6+
- VS Code
- Live Server

Index.html

```
<!DOCTYPE html>
<html><head><title>Palindrome Checker</title><link rel="stylesheet" href="style.css"></head><body><div class="container"><h2>Palindrome Checker</h2><input id="text" type="text" placeholder="Enter text"><button onclick="checkPalindrome()">Check</button><h3 id="result"></h3></div><script src="script.js"></script></body></html>
```

Style.css

```
body{font-family:Arial;background:#eef2ff;display:flex;justify-content:center;align-items:center;height:100vh}.container{background:#fff;padding:30px;border-radius:10px;text-align:center}input,button{padding:10px;margin:10px}
```

Script.js

```
function checkPalindrome() {

    let text = document.getElementById("text").value.trim();

    if (text === "") {

        alert("Please enter a word.");

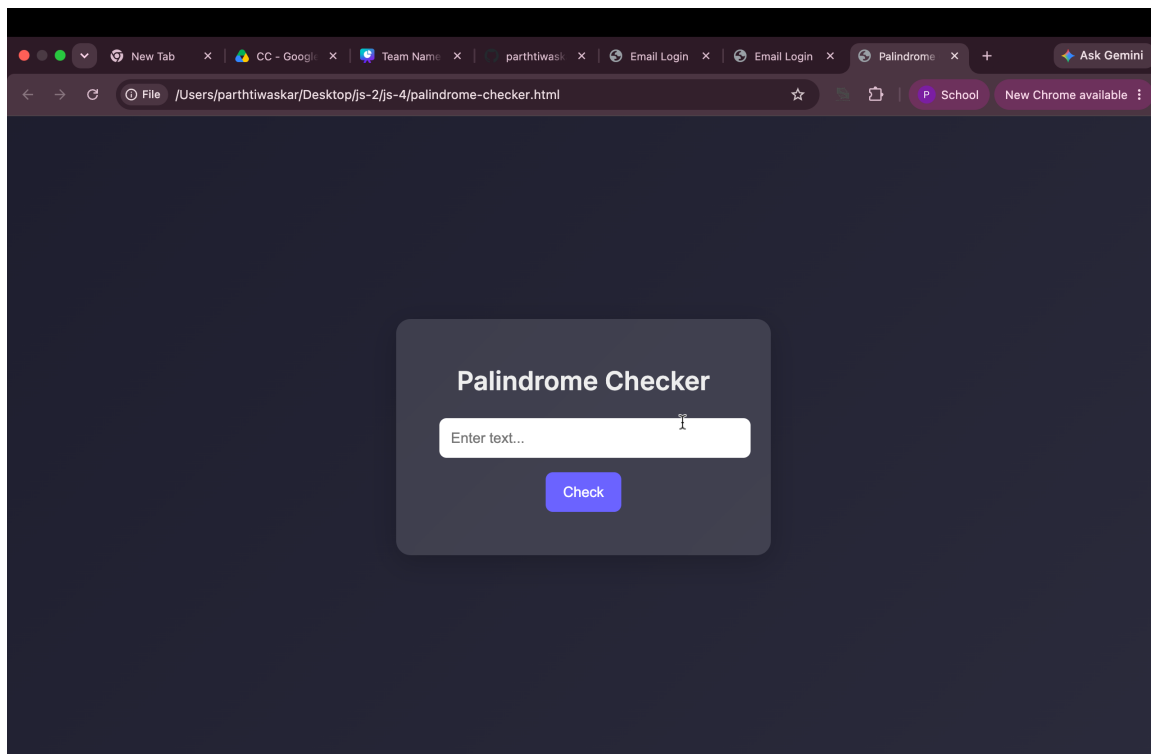
        return;

    }

}
```

```
let original = text.toLowerCase();  
let reversed = original.split("").reverse().join("");  
  
if (original === reversed) {  
    document.getElementById("result").innerHTML =  
        `${text} is a Palindrome `;  
} else {  
    document.getElementById("result").innerHTML =  
        `${text} is NOT a Palindrome `;  
}  
}
```

5. Output



6. Result

The palindrome checker was implemented successfully.

Case Study

Title

Vehicle Registration Number Validation System

Overview

This application validates Indian vehicle registration numbers using Regular Expressions. It accepts a vehicle number and checks whether it follows the format AA00AA0000.

Theory

Regex enables pattern matching. The pattern `^[A-Z]{2}[0-9]{2}[A-Z]{1,2}[0-9]{4}$` validates state code, RTO code, series and vehicle number.

Index.html

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Vehicle Registration Validation</title>
```

```
  <link rel="stylesheet" href="style.css">
```

```
</head>

<body>

<div class="container">

  <h2>Vehicle Registration Validation</h2>

  <input type="text"

    id="vehicleNo"

    placeholder="Enter Vehicle Number">

  <button onclick="validateVehicle()">

    Validate

  </button>

  <h3 id="result"></h3>

</div>

<script src="script.js"></script>

</body>

</html>
```

Style.css

```
body{

  font-family: Arial;

  background:#f5f5f5;
```

```
}
```

```
.container{  
  width:400px;  
  margin:80px auto;  
  background:white;  
  padding:20px;  
  border-radius:10px;  
  text-align:center;  
}
```

```
input{  
  width:90%;  
  padding:10px;  
  margin:15px 0;  
}
```

```
button{  
  padding:10px 20px;  
  cursor:pointer;  
}
```

```
#result{  
  margin-top:20px;
```

```
}Script.js
```

```
function checkPalindrome() {  
  let text = document.getElementById("text").value.trim();
```

```
if (text === "") {  
    alert("Please enter a word.");  
    return;  
}  
  
let original = text.toLowerCase();  
let reversed = original.split("").reverse().join("");  
  
if (original === reversed) {  
    document.getElementById("result").innerHTML =  
        `${text} is a Palindrome `;  
} else {  
    document.getElementById("result").innerHTML =  
        `${text} is NOT a Palindrome `;  
}  
}
```

Output

Result/Conclusion

The system successfully validates Indian vehicle registration numbers using JavaScript, Regex and DOM manipulation.

